ASHA American Speech-Language-Hearing Association

AJSLP

Research Article

# Effectiveness of Constraint-Induced Language Therapy for Aphasia: Evidence From Systematic Reviews and Meta-Analyses

**Anastasia M. Raymer[a] and Jane Roitsch[b]**

[a] Department of Communication Disorders and Special Education, Old Dominion University, Norfolk, VA [b] Department of Communication Disorders, University of Nebraska–Kearney

ARTICLE INFO

Article History:
Received August 11, 2022
Revision received October 3, 2022
Accepted October 28, 2022

Editor-in-Chief: Katherine C. Hustad
Editor: Michael de Riesthal

https://doi.org/10.1044/2022_AJSLP-22-00248

ABSTRACT

**Purpose:** Constraint-induced language therapy (CILT) is an aphasia treatment that incorporates neuroplasticity principles of forced verbal use and high-intensity training to facilitate language recovery in individuals with stroke-induced aphasia (Pulvermüller et al., 2001). The burgeoning CILT literature has led to systematic reviews (SRs) that summarize treatment results. In this project, we appraised the quality and examined findings reported in several SRs to draw conclusions about the effectiveness of CILT.
**Method:** We searched multiple databases for SRs that summarized CILT research for poststroke aphasia. We identified six SRs, among which three summarized findings qualitatively and three included meta-analysis (MA) to quantify results. We rated each SR for methodologic quality using the A MeaSurement Tool to Assess Systematic Reviews (AMSTAR 2; Shea et al., 2017) and extracted findings across the six SRs.
**Results:** Two reviewers reliably applied the AMSTAR 2 to the six SRs. Although the six SRs generally were conducted with satisfactory rigor, each was lacking two or more critical domains. Descriptive summaries in SRs reported positive effects of CILT for language and communication measures. However, the three MAs showed that effects of CILT often did not surpass those of comparison treatments for naming, comprehension, and repetition measures. MA findings were positive in a review that included all research designs and evaluated treatment effects for trained naming items. Generalized CILT effects for standardized language measures were limited in two other MAs.
**Conclusions:** CILT led to improvements in a variety of language and communication measures. When compared with intensive multimodality treatments, CILT effects were similar, suggesting that training intensity may be the potent factor in CILT outcomes. Future SRs should be implemented with increased rigor across quality rating scale domains to increase confidence in conclusions.

Constraint-induced language therapy (CILT) is an aphasia treatment approach that incorporates use-dependent principles to facilitate language recovery in individuals with aphasia following left hemisphere stroke (Pulvermüller et al., 2001). Specifically, CILT engages individuals with aphasia in therapeutic language activities that constrain responses to the verbal modality, that is, forced use. The quantity and quality of verbal productions are shaped and reinforced by the clinician during structured interactions with other persons with aphasia. Importantly, the CILT schedule represents high-intensity treatment. Patients with aphasia participate in as many as 3 hr per day of treatment delivered over a 2- to 3-week period. Since 2001, CILT, sometimes called Intensive Language Action Therapy (ILAT; Difrancesco et al., 2012), has garnered considerable interest in clinical and research circles, including dozens of original research

Correspondence to Anastasia M. Raymer: sraymer@odu.edu. **Publisher Note:** This article is part of the Special Issue: Select Papers From the 51st Clinical Aphasiology Conference. ***Disclosure:*** *The authors have declared that no competing financial or nonfinancial interests existed at the time of publication.*

studies that have examined effects of CILT for aphasia recovery.

CILT contrasts with multimodality forms of aphasia treatment in which communicative success is fostered through various expressive modalities in addition to spoken word production (Pierce et al., 2019). These modalities can include the use of gestures, writing, or drawing to communicate ideas. Multimodality treatment protocols sometimes pair these alternative communication modalities with verbal production to encourage recovery of language functions. For example, Rose, Raymer, et al. (2013) in a systematic review (SR) reported considerable improvements in noun and verb retrieval following verbal + gesture treatments for aphasia. Researchers have examined multimodality treatment delivered at a high treatment intensity and reported recovery of language comprehension and production abilities (Attard et al., 2013; Rose, Attard, et al., 2013).

Remaining abreast of the growing aphasia treatment research literature is a considerable challenge for busy clinicians seeing patients with aphasia in daily practice as demands on time and limited access to publications are often barriers to evidence-based clinical practice (Shrubsole et al., 2019). Research summaries are highly valued documents that provide clinicians with access to synopses of research outcomes on a topic with conclusions about the strength of the overall body of research findings. While dozens of summary approaches exist (Grant & Booth, 2009), SRs represent a research synopsis in which authors conduct a comprehensive literature search and synthesize the resulting body of research to answer clinical questions. While results of SRs typically are descriptive, some include meta-analysis (MA) in which a quantitative synthesis of the research results takes place for common outcome metrics across studies. SRs also typically incorporate an appraisal step in which the original research studies are examined for methodologic quality, so conclusions can be based on the strongest empirical evidence available. Thus, SRs provide clinicians with a concise summary of the effects of a given treatment reported in the literature, often ending with practice guidelines or statements.

In 2008, the first SR of CILT summarized findings of five original studies, reporting positive effects of the approach for language and communication measures as compared with general, less intensive treatment methods (Cherney et al., 2008). Since that time, dozens more original CILT studies have been conducted and several updated SRs have taken place. In contrast to the 2008 SR, the conclusions of recent SRs have been more tenuous, including some MAs with less optimistic conclusions about effects of CILT compared with other intensive treatments (Zhang et al., 2017).

One factor leading to different conclusions across CILT SRs is the rigor of methods employed (Schlosser et al., 2007). To that end, tools have been developed to assist in the appraisal of SR methods and conclusions. In contrast to the well-known PRISMA statement developed to guide *writing* of SRs (Preferred Reporting Items for Systematic Reviews and Meta-Analyses [PRISMA]; Page et al., 2021), researchers within epidemiology developed a critical *appraisal* instrument for SRs called A MeaSurement Tool to Assess Systematic Reviews (AMSTAR 2; Shea et al., 2017). The AMSTAR 2 checklist guides readers of SRs to identify features that should be present to assure rigor and lack of bias in the conduct and conclusions of a review. This includes items such as a comprehensive literature search with defined inclusion and exclusion criteria to identify original studies that address a given clinical question, reliable selection of the research studies and extraction of the resulting data, and careful synthesis and discussion of the research results.

The purpose of the current project was to examine SRs that summarize treatment effects of CILT. Specifically, we asked the following: What is the effect of CILT for improvement of language and communication outcomes in individuals with stroke-induced aphasia? We endeavored to evaluate the weight of the research evidence for CILT treatment outcomes considering a quality appraisal for each of the SRs.

## Method

We searched several databases (PsychInfo, PubMed, SpeechBite, American Speech-Language-Hearing Association Evidence Maps) for SRs published through December 2021 using the search terms "aphasia" and "treatment or intervention or therapy or management" and "systematic review or meta-analysis." We downloaded 143 documents. Among those documents, we reviewed titles and abstracts to select those examining effects of CILT in poststroke aphasia. Through consensus, we identified six SR papers summarizing CILT research, three that provided descriptive SR results (Balardin & Miotto, 2009; Cherney et al., 2008, 2010) and three that included MA to quantify results statistically (Pierce et al., 2019; Wang et al., 2020; Zhang et al., 2017).

To conduct a quality appraisal of each SR, we used the AMSTAR 2 (Shea et al., 2017), a 16-item tool developed within epidemiology to complement the PRISMA writing statement. AMSTAR 2 items evaluate the presence of features deemed essential to the rigor and lack of bias in the SR process, including comprehensive search guided by a clinical question, reliable study selection and data extraction, risk of bias measurement for original studies, rigorous synthesis of results, and lack of conflict of interest. Shea et al. (2017) considered seven of the 16 AMSTAR 2 items as "critical domains" to demonstrate validity and reduce any bias that could be introduced in the SR methods. The AMSTAR 2 items are listed in Table 1, and critical domains are noted with an asterisk.

In the current project, the two authors trained on the use of the AMSTAR 2 by independently reviewing one paper (Cherney et al., 2008), discussing resulting scores, and achieving consensus. The authors then independently coded the other five SRs. Disagreements were discussed to achieve consensus ratings. One examiner (A.R.) then extracted information from each SR pertaining to number of studies examined, research quality rating tool used, types of treatment designs included in each review, number of participants with stroke, aphasia chronicity, type of CILT treatment and comparison treatment, schedule of treatment, and outcomes/conclusions of each review.

## Results

### Critical Appraisal of SRs

The results of the AMSTAR 2 critical appraisal for the six SRs are shown in Table 1. The two authors achieved agreement reliability of 94.8% (91/96) in ratings across reviews. The quality of the SRs on the AMSTAR 2 scale ranged from 1.5 items addressed (Balardin & Miotto, 2009) to 10.5 items included (Wang et al., 2020).

All six SRs noted the source of funding and any conflict of interest involving the review authors (No. 16). Five SRs implemented a research question and inclusion criteria that represented the PICO format (population, intervention, comparison, and outcomes; No. 1). Five SRs also described the included studies in detail (No. 8), typically in a summary table. Importantly, five of the SRs assessed risk of bias of the included studies (No. 9), a critical domain in AMSTAR 2 (Shea et al., 2017). Four of the six SRs addressed the following appraisal items: study selection in duplicate (No. 5), data extraction in duplicate (No. 6), consider risk of bias when interpreting results (No. 13), and explain heterogeneity of results (No. 14). Of the projects that evaluated methodologic quality of the original CILT studies, only Pierce et al. (2019) explicitly summarized evidence relative to quality indicators. For the three reviews that conducted MA in the results, all were noted to use an appropriate combination of statistical findings (No. 11), and two of them considered risk of bias in reporting the results (No. 12).

Five study appraisal items were lacking across all reviews. Of most concern are three of those five studies that are missing critical domains of the AMSTAR 2 to document an unbiased SR process: established protocol along with protocol deviations justified (No. 2), lists of excluded studies with exclusions justified (No. 7), and investigation of publication bias (No. 15). The final two lacking items were that only two reviews justified selection of study designs (No. 3) and none mentioned sources of funding of studies included in the reviews (No. 10).

As noted above, Balardin and Miotto (2009) addressed few of the AMSTAR 2 items, including only one of seven critical domains. The other five SRs, which

**Table 1.** Results of A MeaSurement Tool to Assess Systematic Reviews (AMSTAR 2) checklist items for each constraint-induced language therapy (CILT) systematic review scored as 0–1 following scoring guidelines.

| AMSTAR 2 items | Cherney et al., 2008 SR | Balardin & Miotto, 2009 SR | Cherney et al., 2010 SR | Zhang et al., 2017 SR + MA | Pierce et al., 2019 SR + MA | Wang et al., 2020 SR + MA | No. of 6 reported |
|---|---|---|---|---|---|---|---|
| 1. PICO research questions/inclusion criteria | 1 | 0 | 1 | 1 | 1 | 1 | 5 |
| 2.Established protocol/deviations justified | 0 | .5 | 0 | 0 | 0 | .5 | 2[a] |
| 3. Justify selection of study designs | 0 | 0 | 0 | 1 | 0 | 1 | 2 |
| 4. Comprehensive literature search strategy | 0 | 0 | 0 | .5 | .5 | .5 | 3[a] |
| 5. Study selection in duplicate | 1 | 0 | 1 | 1 | 1 | 0 | 4 |
| 6. Data extraction in duplicate | 1 | 0 | 1 | 1 | 0 | 1 | 4 |
| 7. List of excluded studies/justify exclusions | 0 | 0 | 1 | 0 | 0 | 0 | 1[a] |
| 8. Describe included studies in detail | .5 | .5 | .5 | .5 | 0 | .5 | 5 |
| 9. Assess risk of bias in included studies | 1 | 0 | .5 | 1 | 1 | 1 | 5[a] |
| 10. Sources of funding for included studies | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11. Appropriate statistical combination | NA | NA | NA | 1 | 1 | 1 | 3/3[a] |
| 12. Risk of bias impact on results | NA | NA | NA | 0 | 1 | 1 | 2/3 |
| 13. Risk of bias when interpreting results | 1 | 0 | 1 | 0 | 1 | 1 | 4[a] |
| 14. Explanation for results heterogeneity | 1 | 0 | 0 | 1 | 1 | 1 | 4 |
| 15. Investigation/impact of publication bias | NA | NA | NA | 0 | 0 | 0 | 0[a] |
| 16. Conflict of interest/funding for review | 1 | 1 | 1 | 1 | 1 | 1 | 6 |
| Total number of items addressed | 7.5 | 1.5 | 7 | 9 | 9.5 | 10.5 | |

*Note.* SR = systematic review; MA = meta-analysis; PICO = population, intervention, comparison, and outcomes; NA = not applicable.
[a]Indicates items that are critical domains.

appear acceptable for receiving credit for a majority of the AMSTAR 2 items, did not meet several critical domains. Cherney et al. (2008) met two of seven critical domains. Cherney et al. (2010) and Zhang et al. (2017) addressed three of seven critical domains. Pierce et al. (2019) included information for four of seven critical domains. In addition, Wang et al. (2020) met five of seven critical domains. Thus, all would be considered "critically low" in the AMSTAR 2 confidence scale as each was lacking two or more critical domains. These weaknesses should be taken into account when interpretating the overall findings of the six SRs.

## Review Findings

Characteristics of the six CILT SRs are summarized in Table 2. The number of original studies included in the SRs ranged from five to 24 that incorporated CILT, as some reviews included only randomized controlled trials (RCTs; Wang et al., 2020: 12 studies), whereas others included both group and single participant designs (Pierce et al., 2019: 24 studies). Total participants summarized in the CILT studies ranged from 76 to 493 across the SRs, again influenced by the types of treatment designs included. The phase of aphasia recovery was chronic in three SRs, chronic + subacute + acute in two, and not stated in one. Studies typically included more persons with nonfluent than fluent aphasias and mostly individuals with moderate aphasia.

While each SR summarized findings of CILT research studies, types of comparison treatments varied across the reviews. Some included any comparison treatment, regardless of treatment schedule or response modality (Balardin & Miotto, 2009; Cherney et al., 2008, 2010), whereas others ascertained that the comparison treatment was delivered at the same intensive treatment schedule (Pierce et al., 2019; Wang et al., 2020; Zhang et al., 2017).

A synopsis of findings for language measures (naming, comprehension, repetition, and overall aphasia test batteries) and activity/participation outcomes (discourse, communication logs, and rating scales) are listed in Table 3. The number of studies indicating positive results is noted relative to the number of studies examining that type of outcome measure. Positive CILT findings were reported in the majority of original studies. However, when results were summarized with three MAs that considered mean performance and confidence intervals of CILT relative to control conditions, varied findings resulted.

### Language Outcomes

All six SRs reported findings for comprehensive aphasia test batteries, typically the Aachen Aphasia Test (AAT; Huber et al., 1984) or the Western Aphasia Battery (WAB; Kertesz, 1982). As many as nine of 10 original studies favored CILT relative to a comparison treatment condition (Cherney et al., 2010). In contrast, two of the MAs reported CILT advantages in only one of six studies (Zhang et al., 2017) and two of four studies (Pierce et al., 2019). In a quantitative synthesis, Wang et al. (2020) reported no significant difference for CILT relative to the comparison treatment condition in two studies using the AAT, whereas a significant change was reported in favor of CILT on the WAB for two studies combined.

All six SRs reported findings for naming measures across the reviewed studies, typically the AAT Naming subtest, the WAB Naming subtest, or the Boston Naming Test (Kaplan et al., 2001). As many as five of eight studies reported naming improvements associated with CILT (Cherney et al., 2010). MA findings indicated positive effects of CILT in single-subject research designs examining trained naming stimuli (Pierce et al., 2019). In contrast, generalized naming improvements for *untrained* stimuli were not significant in two MAs examining CILT results for standardized naming tests (Wang et al., 2020; Zhang et al., 2017). A similar pattern was evident in measures of repetition, comprehension, and spontaneous speech. The SRs reported positive CILT effects in as many as nine of 10 studies for comprehension measures, three of four studies for repetition measures, and one of two studies for spontaneous speech outcomes (Cherney et al., 2010). However, MAs indicated no significant CILT effects over comparison treatments for comprehension, repetition, and spontaneous speech analyses (Wang et al., 2020; Zhang et al., 2017). Four of the six SRs also noted maintenance of treatment effects (see Table 3). While as many as four original CILT studies examined maintenance, only two studies documented lasting effects of treatment beyond 1 month posttreatment completion.

## Communication Activity/Participation Measures

Descriptive results of communication activity/participation measures are summarized in five of six SRs (none reported in Wang et al., 2020). While discourse outcomes reportedly favored CILT in six of seven studies (Cherney et al., 2010), Zhang et al. (2017) noted no CILT advantages for discourse outcomes in two studies examined. Communication logs were employed in several studies, favoring CILT relative to comparison conditions in as many as three of four studies (Zhang et al., 2017). While few studies examined outcomes in communication rating scales, Pierce et al. (2019) reported findings favoring CILT in two of three studies. Among the six SRs, the varied communication activity/participation metrics employed were not amenable to quantitative synthesis.

**Table 2.** Characteristics of six systematic reviews (SRs) summarizing constraint-induced language therapy (CILT) research.

| Authors | No. of studies reviewed | Quality rating method | Design of treatment studies | No. of participants | Type of treatment reviewed | Schedule of treatment |
|---|---|---|---|---|---|---|
| Cherney et al., 2008 SR | 5 CILT of 10 total | ASHA levels of evidence | Group and SSD | 90 CILT chronic stroke | CILT/CILTplus vs. comparison tx | 3 hr × 4–5x/wk × 2 wks |
| Balardin & Miotto, 2009 SR | 5 CILT | AANS classif. hierarchy | RCTs and nonrandomized controlled trials; case series | 39 CILT 37 controls chronic stroke | CILT/CILTplus vs. control tx | NS |
| Cherney et al., 2010 SR | 13 new CILT + 5 prior | ASHA levels of evidence | Group & SSD | 100 CILT chronic stroke | CILT with modifications | 15–36 hr |
| Zhang et al., 2017 SR + MA | 8 CILT | Cochrane | RCTs | 132 CILT 140 controls Stroke (5 chronic, 1 subacute, 2 acute studies) | CILT & ILAT vs. comparison tx | 20–40 hr |
| Pierce et al., 2019 SR + MA | 24 CILT of 60 total | PEDro, RoBiNT | Group & SSD | 254 CILT 186 controls stroke TPO NS | CILT vs. multimodal tx | NS |
| Wang et al., 2020 SR + MA | 12 CILT | Cochrane | RCTs | 493 CILT + controls stroke 7 chronic 2 subacute 3 acute | CILT vs. comparison tx | NS |

*Note.* ASHA = American Speech-Language-Hearing Association; SSD = single-subject design; tx = treatment; ANNS = American Association of Neurologic Surgeons; RCT = randomized controlled trial; MA = meta-analysis; ILAT = Intensive Language Action Therapy; PEDro = Physiotherapy Evidence Database; RoBiNT = Risk of Bias in N-of-1 Trials; TPO NS = time post onset not stated.

**Table 3.** Number of studies reporting positive outcomes of constraint-induced language therapy (CILT) compared with control treatment for language and communication activity/participation measures in six systematic reviews (SRs; $n$ = 3 descriptive SRs and $n$ = 3 SR with meta-analysis).

| Authors | Language outcomes | Communication activity/ participation | Maintenance |
|---|---|---|---|
| Cherney et al., 2008<br>SR | Aphasia battery: 4/4<br>Naming: 2/3<br>Repetition: 1/1<br>Comprehension: 3/3 | Discourse tasks: 0/1<br>Communication log: 2/2<br>Rating scale: 1/1 | CILT effects maintained in 2/2 studies |
| Balardin & Miotto, 2009<br>SR | Aphasia battery: 4/4<br>Naming: 4/4<br>Repetition: 2/3<br>Comprehension: 4/4 | Discourse tasks: 2/2<br>Communication log: 2/3<br>Rating scale: 1/1 | CILT effects maintained in 2/2 studies |
| Cherney et al., 2010<br>SR | Aphasia battery: 9/10<br>Naming: 5/8<br>Repetition: 3/4<br>Comprehension: 9/10<br>Spontaneous speech: 1/2 | Discourse tasks: 6/7<br>Communication log: 2/2<br>Rating scale: 1/1 | CILT effects maintained in 2/4 studies |
| Zhang et al., 2017<br>SR + MA | Aphasia battery: 1/6<br>Naming: 2/3; meta-analysis MD = 3.97, N.S.<br>Repetition: 1/2; meta-analysis MD = .08, N.S.<br>Comprehension: 2/2; meta-analysis MD = −4.34, N.S. | Discourse tasks: 0/2<br>Communication log: 3/4<br>Rating scale: 0/1 | CILT effects maintained in 1/1 |
| Pierce et al., 2019<br>SR + MA | Aphasia battery: 2/4<br>Naming: 4/5; meta-analysis Tau-U = 0.374<br>Repetition: 2/2<br>Comprehension: 1/2 | Discourse tasks: 1/2<br>Communication log: 1/1<br>Rating scale: 2/3 | |
| Wang et al., 2020<br>SR + MA | Aphasia battery: meta-analysis AAT MD = −1.11, N.S.; WAB MD = 1.23, $p$ < .01<br>Naming: 2/3; meta-analysis BNT MD = −1.79, N.S.; AAT MD = −1.11, N.S.; WAB MD = 1.85, $p$ < .001<br>Repetition: 1/3; meta-analysis AAT MD = 0.08, N.S.<br>Spontaneous speech: meta-analysis WAB MD = 0.51, N.S.<br>Comprehension: 2/3; meta-analysis Token test MD = −0.60, N.S.; AAT MD = −4.34, N.S. | | |

*Note.* MA = meta-analysis; MD = mean difference; N.S. = not significant; AAT = Aachen Aphasia Test; WAB = Western Aphasia Battery; BNT = Boston Naming Test.

## Discussion

In this project, we identified six SRs summarizing the burgeoning CILT treatment literature. In those six SRs, we listed every individual study and identified 42 different original CILT studies, which were included for review. Among the 42 were 12 randomized controlled trials (RCTs), which are often considered the gold standard in treatment research (Robey, 2004). It would be an enormous undertaking for clinicians to remain up-to-date and informed of such an expanding literature. Thus, access to SRs summarizing CILT research findings represents a valuable resource to clinicians to guide clinical decision making and appropriate patient candidacy for CILT. To the question whether CILT is an effective treatment to promote aphasia recovery following left hemisphere stroke, a close examination of this set of reviews resulted in varied answers depending upon *how* the SRs were conducted and the nature of the outcomes measured.

One impactful variation in SRs is the type of research designs that was included in the review. Several CILT SRs incorporated all treatment research designs, including case series and single-subject designs (Balardin & Miotto, 2009; Cherney et al., 2008, 2010; Pierce et al., 2019). When all treatment research designs were included, SR conclusions were often optimistic about the effects of CILT for language and communication outcomes. The quantitative synthesis by Pierce et al. (2019) showed powerful CILT effects for trained naming stimuli. The two SRs that summarized only RCTs of CILT (Wang et al., 2020; Zhang et al., 2017) conducted MAs and reported limited positive outcomes for standardized language measures available for quantitative synthesis. Wang et al. (2020) reported significant changes in the MA for the WAB using data from only two original CILT studies, and no significant changes for the AAT scores reported in two original CILT studies. These observations suggest that while CILT treatment effects may be potent for trained language stimuli, the effects may be no more advantageous relative to comparison multimodality treatments when examining generalization to untrained language measures. The variety of activity/participation measures has precluded quantitative analysis, but positive outcomes have been reported across studies, particularly for communication logs (Zhang et al., 2017).

Another variation in conclusions of SRs encompasses the comparison treatment conditions considered as part of the clinical questions posed. CILT incorporates two unique principles of neuroplasticity within the treatment protocol: forced use of verbal responses and intensive treatment schedule. Earlier SRs considered CILT effects relative to any comparison treatments, including traditional multimodality treatments or those delivered at a less intensive schedule (e.g., Cherney et al., 2008, 2010). Conclusions suggested that CILT led to superior outcomes for many language and communication activity/participation measures relative to traditional, less intensive treatment protocols. Recent SRs systematically assessed CILT effects relative to comparison treatments incorporating unconstrained multimodality treatment options where intensity was held constant (Pierce et al., 2019; Zhang et al., 2017). Conclusions of those reviews indicated that CILT effects for language and communication outcomes are often no different than effects of multimodality treatments delivered at a comparable intensity schedule. This observation suggests that intensity may be the more potent factor in CILT outcomes compared with constraint to use verbal responses. CILT studies that systematically vary both treatment intensity and verbal constraint are needed to fortify this conclusion.

### Methodological Considerations

Another observation that we made in this project examining six SRs of CILT research is that the methodological quality of SRs may be undermined because key elements are lacking to assure rigor and lack of bias in the review, despite the guidance provided by publications and appraisal tools promulgated through epidemiology (e.g., PRISMA statement, Page et al., 2021; AMSTAR 2, Shea et al., 2017) and speech-language pathology (e.g., Patterson & Raymer, 2022). All six SRs in this project were lacking AMSTAR 2 critical domains, thereby conclusions drawn must be considered in light of potential bias in the reviews. Research groups conducting SRs should incorporate processes and descriptions that ensure validity and lack of bias in the review process to increase confidence in conclusions of the reviews. Among the six CILT SRs, little information was provided about an established protocol being followed and any justifications for deviations from that protocol. While each SR attempted to conduct a comprehensive literature search, details of those searches (such as language restrictions and search terms) were often lacking, which undermines confidence that all studies on a given topic were captured in the search and the process could be replicated in future reviews. Support for this comment comes by noting that each of the SRs encompassed varied subsets of the 42 original CILT studies listed across the six reviews. Only one SR provided a list of excluded studies with justifications for exclusion (Cherney et al., 2010), another critical domain that hedges against bias in the selection process that could favor selection of studies with positive findings. Finally, publication bias is a critical domain of the AMSTAR 2 that was not considered in any of the six SRs. A search for gray literature that includes unpublished findings is a mechanism to counter the risk of publication bias. As quantitative synthesis of CILT research findings improves, consideration should be given to the preponderance of the evidence and whether studies with negative findings may be unpublished and missing from the review, thereby unintentionally exaggerating interpretations of the positive effects of CILT. To the extent that CILT studies are replicating positive effects in multiple research labs around the world, however, publication bias is a less concerning factor (Shea et al., 2017).

## Limitations, Future Directions, and Clinical Implications

As with most research studies, we must consider some limitations of our project that may affect our findings and interpretations. Although the search string we utilized was formulated with care, it is still possible that some terms were missing such that other SRs were not identified in the search process. We searched only published academic papers from a select range of years since publication of the original CILT paper in 2001 (Pulvermüller et al., 2001) to provide current and evidence-based information. Thus, our results may have missed nonacademic works that have not been listed in scholarly databases.

The study selection, inclusion process, and appraisal ratings were performed by two reviewers, one of whom is a co-author of two of the SRs (Cherney et al., 2008, 2010). Despite that shortcoming, the high reliability of AMSTAR 2 ratings increases confidence that bias was avoided in the SR appraisal process.

The varied designs of the papers reviewed (e.g., single-subject and group designs) across SRs may afford less-than-optimal results comparisons and MA limited to only a few original CILT studies. However, some outcomes, such as communication activity/participation, are more amenable to examination in single-subject treatment designs. Ultimately, rigorous SRs depend on the existence of high-quality original research, particularly RCTs, which incorporate common outcome measures to facilitate future MAs (Wallace et al., 2019).

This work also has implications for clinicians using SRs to guide clinical decision making. We demonstrate that the AMSTAR 2 (Shea et al., 2017) is an efficient tool for evaluating the quality of SRs. Generally, SRs of high quality would instill greater confidence in the results and should be given higher weight to guide choices made in daily clinical practice. The AMSTAR 2 sets the bar high as seven items on the checklist are considered critical domains to assure that reviewers avoid any bias and engage in a rigorous review process that finds the breadth of the literature to address a clinical question. Unfortunately, we found that although most of the CILT SRs appeared to be conducted with reasonable rigor by addressing the majority of AMSTAR 2 items, all were lacking more than one critical domain, which undermines to some extent the confidence we may have in the review interpretations. Following the AMSTAR 2 guidance, we cannot advocate that any one of the CILT SRs garners greater confidence and weight than another in guiding clinical practice.

Mindful of the limitations of the SR quality, this summary of six SRs of CILT suggests that CILT is an effective treatment approach with outcomes comparable with alternative multimodality treatments to improve language and communication of individuals with stroke-induced aphasia. That is, both forced verbal production and multimodality methods have positive outcomes for individuals with aphasia if the treatments are delivered with sufficient intensity in terms of dose frequency per week and cumulative amount of training over time. While the assumption would be that this positive conclusion would be based on a growing number of CILT research studies over time, the conclusion comes from MA of only two to three RCTs that have manipulated one factor, constraint versus multimodality, while controlling the other factor, intensity of training, to examine outcomes for common language measures (Wang et al., 2020; Zhang et al., 2017). Thus, future high-quality RCTs are needed in which the key factors of CILT, constraint, and intensity are carefully manipulated and common outcome metrics are examined (Wallace et al., 2019).

Recognizing the importance of aphasia treatment delivered in an intensive manner, clinical groups have developed Intensive Comprehensive Aphasia Programs (ICAPs) where individual and group language and communication activities take place within a 2- to 3-hr daily treatment schedule for several weeks (Rose, Attard, et al., 2013). The question remains whether funding sources are amenable to support treatment delivered at such a condensed high-intensity schedule. Recent analyses indicate that, while costly overall, the cost–benefit of such an intensive aphasia treatment program appears to improve over time (Boyer et al., 2022), although additional analyses of cost effectiveness are needed.

# Conclusions

The CILT treatment literature continues to grow, and several international research groups have summarized that expansive literature in six SRs conducted over recent years. Early conclusions about the positive effects of CILT for recovery of language and communication abilities have given way to more measured conclusions based on quantitative synthesis of findings. Closer examination of those findings suggests that, indeed, CILT is an effective aphasia treatment approach, with outcomes that match other unconstrained multimodality treatments delivered at a comparable high-intensity schedule. Future high-quality experimental studies including more RCTs that systematically manipulate both verbal constraint and treatment intensity are needed as well as cost–benefit analyses that determine how intensive treatments can be delivered and funded to maximize patient outcomes.

## Data Availability Statement

All data for this project are provided in the included tables.

## References


**Attard, M. C., Rose, M. L., & Lanyon, L. E.** (2013). The comparative effects of multi-modality aphasia therapy and constraint-induced aphasia therapy-plus for severe chronic Broca's aphasia: An in-depth pilot study. *Aphasiology, 27*(1), 80–111. http://doi.org/10.1080/02687038.2012.725242

**Balardin, J. B., & Miotto, E. C.** (2009). A review of constraint-induced therapy applied to aphasia rehabilitation in stroke patients. *Dementia Neuropsychology, 3*(4), 275–282. https://doi.org/10.1590/S1980-57642009DN30400003

**Boyer, N., Jordan, N., & Cherney, L. R.** (2022). Implementation cost analysis of an intensive comprehensive aphasia program. *Archives of Physical Medicine and Rehabilitation, 103*(7), S215–S221. https://doi.org/10.1016/j.apmr.2020.09.398

**Cherney, L., Patterson, J., Raymer, A., Frymark, T., & Schooling, T.** (2010, October). *Updated evidence-based systematic review: Effects of intensity of treatment and constraint-induced language therapy for individuals with stroke-induced aphasia*. Report of the ASHA National Center for Evidence-Based Practice in Communication Disorders

**Cherney, L. R., Patterson, J. P., Raymer, A., Frymark, T., & Schooling, T.** (2008). Evidence-based systematic review: Effects of intensity of treatment and constraint-induced language therapy for individuals with stroke-induced aphasia. *Journal of Speech, Language, and Hearing Research, 51*(5), 1282–1299. https://doi.org/10.1044/1092-4388(2008/07-0206)

**Difrancesco, S., Pulvermuller, F., & Mohr, B.** (2012). Intensive language-action therapy (ILAT): The methods. *Aphasiology, 26*(11), 1317–1351. https://doi.org/10.1080/02687038.2012.705815

**Grant, M. J., & Booth, A.** (2009). A typology of reviews: An analysis of 14 review types and associated methodologies. *Health Information and Libraries Journal, 26*(2), 91–108. https://doi.org/10.1111/j.1471-1842.2009.00848.x

**Huber, H. P., Poeck, K., & Williams, K.** (1984). The Aachen aphasia test. In F. C. Rose (Ed.), *Progress in aphasiology* (pp. 291–303). Raven Press.

**Kaplan, E., Goodglass, H., & Weintraub, S.** (2001). *Boston Naming Test*. Lea & Febiger.

**Kertesz, A.** (1982). *Western Aphasia Battery*. Grune & Stratton.

**Page, M. J., Moher, D., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., . . . McKenzie, J.** (2021). PRISMA 2020 explanation and elaboration: Updated guidance and exemplars for reporting systematic reviews. *British Medical Journal, 372,* n160. https://doi.org/10.1136/bmj.n160

**Patterson, J. P., & Raymer, A. M.** (2022). Applying appraisal tools in aphasia systematic reviews: A tutorial. *American Journal of Speech-Language Pathology, 31*(5S), 2291–2300. https://doi.org/10.1044/2022_AJSLP-21-00288

**Pierce, J. E., Menahemi-Falkov, M., O'Halloran, R., Togher, L., & Rose, M. L.** (2019). Constraint and multimodal approaches to therapy for chronic aphasia: A systematic review and meta-analysis. *Neuropsychological Rehabilitation, 29*(7), 1005–1041. https://doi.org/10.1080/09602011.2017.1365730

**Pulvermüller, F., Neininger, B., Elbert, T., Mohr, B., Rockstroh, B., Koebbel, P., & Taub, E.** (2001). Constraint-induced therapy of chronic aphasia after stroke. *Stroke, 32*(7), 1621–1626. https://doi.org/10.1161/01.str.32.7.1621

**Robey, R. R.** (2004). A five-phase model for clinical outcome research. *Journal of Communication Disorders, 37*(5), 401–411. https://doi/org/10.1016/j.jcomdis.2004.04.003

**Rose, M. L., Attard, M. C., Mok, Z., Lanyon, L. E., & Foster, A. M.** (2013). Multi-modality aphasia therapy is as efficacious as a constraint-induced aphasia therapy for chronic aphasia: A phase 1 study. *Aphasiology, 27*(8), 938–971. https://doi/10.1080/02687038.2013.810329

**Rose, M. L., Raymer, A. M., Lanyon, L. E., & Attard, M. C.** (2013). A systematic review of gesture treatments for post-stroke aphasia. *Aphasiology, 27*(9), 1090–1127. https://doi.org/10.1080/02687038.2013.805726

**Schlosser, R. W., Wendt, O., & Sigafoos, J.** (2007). Not all systematic reviews are created equal: Considerations for appraisal. *Evidence Based Communication Assessment and Intervention, 1*(3), 138–150. https://doi.org/10.1080/17489530701560831

**Shea, B. J., Reeves, B. C., Wells, G., Thuku, M., Hamel, C., Moran, J., Moher, D., Tugwell, P., Welch, V., Kristjansson, E., & Henry, D. A.** (2017). AMSTAR 2: A critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both. *British Medical Journal, 358*(358), j4008. https://doi.org/10.1136/bmj.j4008

**Shrubsole, K., Worrall, L., Power, E., & O'Connor, D. A.** (2019). Barriers and facilitators to meeting aphasia guideline recommendations: What factors influence speech pathologists' practice? *Disability and Rehabilitation, 41*(13), 1596–1607. https://doi.org/10.1080/09638288.2018.1432706

**Wallace, S. J., Worrall, L., Rose, T., Le Dorze, G., Breitenstein, C., Hilari, K., Babbitt, E., Bose, A., Brady, M., Cherney, L. R., Copland, D., Cruice, M., Enderby, P., Hersh, D., Howe, T., Kelly, H., Kiran, S., Laska, A. C., Marshall, J., & Webster, J.** (2019). A core outcome set for aphasia treatment research: The ROMA consensus statement. *International Journal of Stroke, 14*(2), 180–185. https://doi.org/10.1177/1747493018806200

**Wang, G., Ge, L., Zheng, Q., Huang, P., & Xiang, J.** (2020). Constraint-induced aphasia therapy for patients with aphasia: A systematic review. *International Journal of Nursing Sciences, 7*(3), 349–358. https://doi.org/10.1016/j.ijnss.2020.05.005

**Zhang, J., Yu, J., Bao, Y., Zie, Q., Xu, Y., Zhang, J., & Wang, P.** (2017). Constraint-induced aphasia therapy in post-stroke aphasia rehabilitation: A systematic review and meta-analysis of randomized controlled trials. *PLOS ONE, 12*(8), Article e0183349. https://doi.org/10.1371/journal.pone.0183349

Color legend
========================================

| | | |
|---|---|---|
| claim | #a8c8ff | |
| confounders | #cccccc | |
| imaging_details | #d9b3e6 | |
| limitation | #ffb3b3 | |
| method | #fff4a3 | |
| region_definition | #d9b3e6 | |
| sample | #a8e6a3 | |
| statistics | #ffcc99 | |

Generated by aphasia-kb/annotate_paper.py